

At the body's place we draw the tangent PRZ , the forceless straight path the body would keep without any centripetal pull — this is the heart of the deviation argument. Through the force centre we draw the chord PV , and through V the diameter VA , joining AP . Because VA is a diameter, the angle VPA inscribed in the semi-circle is a right angle, and this single fact seeds the chain of similar triangles that will follow. The chord PV through S is the intrinsic length that will ultimately govern the whole force law.